

**Bhuvaneshvari S 250701111-2025-CSE** ▾**B2****Started on** Monday, 24 August 2026, 10:51 AM**State** Finished**Completed on** Monday, 24 August 2026, 10:52 AM**Time taken** 1 min 11 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int countZeroes(int arr[], int low, int high) {
4      if (low > high)
5          return 0;
6
7      if (arr[low] == 0)
8          return high - low + 1;
9
10     if (arr[high] == 1)
11         return 0;
12
13     int mid = (low + high) / 2;
14
15     return countZeroes(arr, mid + 1, high);
16 }
17
18 int main() {
19     int m;
20     scanf("%d", &m);
21
22     int arr[m];
23
24     for (int i = 0; i < m; i++)
25         scanf("%d", &arr[i]);
26
27     printf("%d\n", countZeroes(arr, 0, m - 1));
28
29     return 0;
30 }
```

	Input	Expected	Got	
✓	5	2	2	✓
	1			
	1			
	1			
	0			
	0			

	Input	Expected	Got	
✓	10 1 1 1 1 1 1 1 1 1 1 1 1	0	0	✓
✓	8 0 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

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**Bhuvaneshvari S 250701111-2025-CSE** ▾**B2****Started on** Monday, 24 August 2026, 10:53 AM**State** Finished**Completed on** Monday, 24 August 2026, 10:53 AM**Time taken** 32 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than $\lfloor n / 2 \rfloor$ times. You may assume that the majority element always exists in the array.

Example 1:

Input: `nums = [3,2,3]`

Output: 3

Example 2:

Input: `nums = [2,2,1,1,1,2,2]`

Output: 2

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int majorityElement(int nums[], int low, int high) {
4      if (low == high)
5          return nums[low];
6
7      int mid = (low + high) / 2;
8
9      int left = majorityElement(nums, low, mid);
10     int right = majorityElement(nums, mid + 1, high);
11
12     if (left == right)
13         return left;
14
15     int leftCount = 0, rightCount = 0;
16
17     for (int i = low; i <= high; i++) {
18         if (nums[i] == left)
19             leftCount++;
20         if (nums[i] == right)
21             rightCount++;
22     }
23
24     return leftCount > rightCount ? left : right;
25 }
26

```

```
27 int main() {  
28     int n;  
29     scanf("%d", &n);  
30  
31     int nums[n];  
32  
33     for (int i = 0; i < n; i++)  
34         scanf("%d", &nums[i]);  
35  
36     printf("%d\n", majorityElement(nums, 0, n - 1));  
37  
38     return 0;  
39 }
```

	Input	Expected	Got	
✓	3 3 2 3	3	3	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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**Bhuvaneshvari S 250701111-2025-CSE** ▾**B2****Started on** Monday, 24 August 2026, 10:53 AM**State** Finished**Completed on** Monday, 24 August 2026, 10:55 AM**Time taken** 1 min 26 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int floorValue(int arr[], int low, int high, int x) {
4     if (low > high)
5         return -1;
6
7     int mid = (low + high) / 2;
8
9     if (arr[mid] == x)
10        return arr[mid];
11
12    if (arr[mid] < x) {
13        int result = floorValue(arr, mid + 1, high, x);
14        if (result == -1)
15            return arr[mid];
16        return result;
17    }
18
19    return floorValue(arr, low, mid - 1, x);
20 }
21
22 int main() {
23     int n, x;
24     scanf("%d", &n);
25
26     int arr[n];
27
28     for (int i = 0; i < n; i++)
29         scanf("%d", &arr[i]);
30
31     scanf("%d", &x);
32
33     printf("%d\n", floorValue(arr, 0, n - 1, x));
34
35     return 0;
36 }
```



	Input	Expected	Got	
✓	6 1 2 8 10 12 19 5	2	2	✓
✓	5 10 22 85 108 129 100	85	85	✓
✓	7 3 5 7 9 11 13 15 10	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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**Bhuvaneshvari S 250701111-2025-CSE** ▾**B2****Started on** Monday, 24 August 2026, 10:55 AM**State** Finished**Completed on** Monday, 24 August 2026, 10:55 AM**Time taken** 29 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  void findPair(int arr[], int left, int right, int x) {
4      if (left >= right) {
5          printf("No\n");
6          return;
7      }
8
9      int sum = arr[left] + arr[right];
10
11     if (sum == x) {
12         printf("%d\n%d\n", arr[left], arr[right]);
13         return;
14     }
15
16     if (sum < x)
17         findPair(arr, left + 1, right, x);
18     else
19         findPair(arr, left, right - 1, x);
20 }
21
22 int main() {
23     int n, x;
24     scanf("%d", &n);
25
26     int arr[n];
27
28     for (int i = 0; i < n; i++)
29         scanf("%d", &arr[i]);
30
31     scanf("%d", &x);
32
33     findPair(arr, 0, n - 1, x);
34
35     return 0;
36 }
```

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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**Bhuvaneshvari S 250701111-2025-CSE** ▾**B2****Started on** Monday, 24 August 2026, 10:56 AM**State** Finished**Completed on** Monday, 24 August 2026, 10:56 AM**Time taken** 33 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```

1  #include <stdio.h>
2
3  void quickSort(int arr[], int low, int high) {
4      if (low >= high)
5          return;
6
7      int i = low, j = high;
8      int pivot = arr[(low + high) / 2];
9
10     while (i <= j) {
11         while (arr[i] < pivot)
12             i++;
13
14         while (arr[j] > pivot)
15             j--;
16
17         if (i <= j) {
18             int temp = arr[i];
19             arr[i] = arr[j];
20             arr[j] = temp;
21             i++;
22             j--;
23         }
24     }
25
26     quickSort(arr, low, j);
27     quickSort(arr, i, high);
28 }
29
30 int main() {
31     int n;
32     scanf("%d", &n);
33
34     int arr[n];
35
36     for (int i = 0; i < n; i++)
37         scanf("%d", &arr[i]);
38
39     quickSort(arr, 0, n - 1);
40
41     for (int i = 0; i < n; i++)
42         printf("%d ", arr[i]);
43
44     return 0;
45 }
```

	Input	Expected	Got	
✓	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✓
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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